

Markscheme

Mathematics: analysis and approaches
Practice question bank

99. (a)

It can be proven by induction that $n! > 2^n$ for every integer $n \geq 4$.

Using this result, find the smallest value of n (with $n \geq 4$) for which $n! - 2^n > 1000$.

check each case: $n = 4 : 8, n = 5 : 88, n = 6 : 656, n = 7 : 4912$

(M1) A1 A1

the first value exceeding 1000 occurs at $n = 7$

A1

[6 marks]